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Testing-Priority Score — Methodology

A transparent heuristic for prioritizing which registered cooling towers to test or inspect next. It is not a prediction that any building has Legionella.

What this is — and what it is not

Is: a way to rank registered cooling-tower buildings by public *compliance and monitoring* signals, so limited inspection/testing effort can focus on the biggest **blind spots** first.

Is not: a risk model for *Legionella* presence. The factors that actually drive growth in a cooling tower — water temperature, biocide dosing and lapses, stagnation, biofilm, tower make/model — are **not in any public dataset**, and the building-level model these signals come from is weak (area-under-curve ≈ 0.70). A high score means "under-monitored, with a history worth a look," not "dangerous."

Inputs (all public — NYC Open Data)

- **Cooling Tower Registrations** (`y4fw-iqfr`) — tower counts and Legionella water-sample dates.
- **Cooling Tower Inspection Results** (`f9wb-g8mb`) — inspection dates and violations, including water-quality / Legionella-specific violations.

All signals are computed **as-of before the investigation window** (capped 2026-07-01) so the outbreak's own sampling can't leak into the score.

The score

Each building is scored on four factors. Every factor is converted to a **0-1 percentile** within the population being ranked (so the score is relative, not an

absolute risk), then combined as a weighted sum:

Factor	What it captures	Weight
Overdue sampling — days since last Legionella test (never-sampled = worst)	monitoring blind spot	0.35
Water-quality / Legionella violations (all-time count)	documented past water-management problems	0.30
Inspection gap — days since last inspection (never = worst)	oversight blind spot	0.20
Tower count	consequence/exposure if it <i>did</i> bloom (bigger plume)	0.15

```
priority = 0.35·overdue + 0.30·violations + 0.20·inspection_gap + 0.15·towers
```

Buildings are ranked high→low. Missing "days since" values (a building never sampled or never inspected) are treated as the worst case — those are exactly the blind spots the score is meant to surface. Weights are a deliberate, editable choice, not a fitted result; anyone can re-weight to match their priorities.

How to read a ranking

- A top rank usually means **overdue or never-tested**, *and* carrying a water-quality violation history — a legitimate reason to **look**, not proof of anything.
- The order is a **triage aid**, not a danger ranking. Two buildings a few places apart are effectively tied given the underlying uncertainty.
- Because the signals are largely *detection/compliance*, the score will tend to surface diligent-but-overdue buildings and larger systems — read it as "where monitoring has gaps," and pair it with actual testing before drawing conclusions.

Limitations

- No public dataset contains the variables that determine *Legionella* growth, so this cannot be a true risk model.

- The parent case-control model is underpowered (only ~74 outbreak cases) and its associations are weak and largely confounded by building size.
- Percentiles are **relative to the group scored** — a "high" score in one ZIP is not comparable to another area with different testing norms.
- Association $\neq$ causation; a score is a prompt to investigate, nothing more.

Reproducibility

Compute the four factors from the two registration/inspection datasets, percentile-rank each within your chosen population (a ZIP, a borough, citywide), apply the weights above, and sort. Any spreadsheet or a few lines of Python reproduce it.

Not an official record. This methodology is an independent analytical aid, not affiliated with the NYC Health Department, and produces no determination about any specific building. For official information, see nyc.gov/health or call 311.